Project Title

Automated Project Task Planning and Scheduling

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

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ABSTRACT

The **Automated Project Task Planning and Scheduling** system is designed to organize and schedule project activities while satisfying dependencies and precedence constraints. The system represents tasks as vertices and their dependencies as directed edges in a graph. **Topological Sorting** is used to determine a valid execution order in which every task is performed only after its required dependencies are completed. Task durations are considered to calculate completion times and determine the **Critical Path** and critical tasks that influence the overall project duration. The system also detects circular dependencies that make scheduling impossible. By combining graph-based scheduling and optimization techniques, the proposed system provides an efficient execution schedule and helps determine the minimum project completion time. It is designed to handle projects containing **up to 1000 tasks**, demonstrating the practical application of Data Structures and Algorithms in project management.

Signature of the Guide